MIL-C-45167C
20 April 1988
SUPERSEDING
MIL-C-45167B
10 February 1976

MILITARY SPECIFICATION

CAPS, RADIATOR: PRESSURE RELIEVING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 2 1/2-inch and a 4 3/4-inch radiator pressure cap with nominal pressure ratings of 7 pound per square inch (psi). Caps covered by this specification are designed for use with industrial and automotive internal-combustion engine radiators (see 6.1).

1.2 Classification. Caps covered by this specification shall be one of the following sizes, as specified (see 6.2):

Size	
A	- 2 1/2-inch radiator pressure cap.
B	- 4 3/4-inch radiator pressure cap.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2930
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:29Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS
FEDERAL

O-A-548	- Antifreeze/Coolant, Engine, Ethylene Glycol, Inhibited, Concentrated.

MILITARY

MIL-A-11755	- Antifreeze, Arctic Type.
MIL-A-53009	- Additive, Antifreeeze Extender, Liquid Cooling System.

STANDARDS
FEDERAL

FED-STD-601	- Rubber, Sampling and Testing.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification, Marking of US Military Property.
MIL-STD-193	- Painting Procedures, Tactical Vehicles.
MIL-STD-45662	- Calibration Systems Requirements.
MS35840	- Cap, Filler Opening: Radiator, 7 PSI Pressure, 2-1/2 inch.
MS35841	- Filler Neck: Radiator (7 PSI).
MS51062	- Cap, Filler Opening: Radiator, Pressure, 7 PSI Assembly.
MS51063	- Neck, Filler, Radiator.

(Copies of specifications and standards required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:29Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B117 - Salt Spray (Fog) Testing, Method of.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish caps, of the specified size, which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the units to be furnished to the Government. All subsequent caps delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein and in referenced specifications and standards (see 4.8.1).

3.2.1 Finish. Cap surfaces shall be cleaned, primed and finished, as applicable in accordance with MIL-STD-193 (see 4.8.1 and 4.8.2).

3.2.2 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.3.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:29Z
Check the source to verify that this is the current version before use.

3.3 Design and construction. Caps shall be fabricated in accordance with the applicable MS specified below (see 4.8.2):

Size A - MS35840
Size B - MS51062

3.3.1 Compatibility with standard filler necks.

3.3.1.1 Type A. Caps shall connect to standard filler neck conforming to MS35841 (see 4.8.2).

3.3.1.2 Type B. Caps shall connect to standard filler neck conforming to MS51063 (see 4.8.2).

3.4 Performance. Caps shall meet the requirements specified below when coolant conforming to O-A-548, MIL-A-53009 or MIL-A-11755 is used.

3.4.1 Fluid resistance. Gaskets shall show no evidence of deterioration when exposed to coolants. Gaskets, immersed in coolant for 158 hours, shall not change in volume more than 3 percent (%). Coolant shall not be discolored and there shall be no deterioration of gasket material due to flaking (see 4.8.3).

3.4.2 Corrosion resistance. After being subjected to a salt spray test for 96 hours, the caps shall subsequently conform to 3.4.4, 3.4.4.1 and 3.4.5 (see 4.8.4).

3.4.3 Extreme low temperature. Contacting the stop on the filler neck at -65 degrees Fahrenheit (°F) for size A, shall require not more than 50 pound-inches (lb in) of torque and size B, shall require not more than 125 lb in of torque. A removal check at -65°F shall be made to determine useability of arctic mittens (see 4.8.5).

3.4.4 Pressure valve. The pressure valve shall relieve at no less than 6.5 psi and no more than 8 psi pressure (see 4.8.6).

3.4.4.1 Leakage. Pressure valves shall evidence no leakage at pressures up to 6.25 psi (see 4.8.2 and 4.8.6.1).

3.4.5 Vacuum valve. The vacuum valve shall relieve at less than 1 psi vacuum (see 4.8.7).

3.5 Identification marking. Caps shall be marked in accordance with the applicable MS and shall be in accordance with MIL-STD-130 (see 4.8.2).

3.6 Workmanship. Workmanship shall be such as to provide a cap free from defects such as: cracked, flaking or damaged gasket, dirty internal parts, defective finish, and sharp edges or burrs (see 4.8.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:29Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10% of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspections:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
    1. Examination (see 4.5.2).
    2. Tests (see 4.5.3).
c. Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 73 $\pm$ 18°F
b. Barometric pressure $28.5 \begin{smallmatrix} +2 \\ -3 \end{smallmatrix}$ inches (in) mercury (Hg)
c. Relative humidity 50 $\pm$ 30%

4.3.1 Cleanliness. Prior to all testing, all internal parts shall be cleaned, removing all foreign matter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:29Z
Check the source to verify that this is the current version before use.

4.3.2 Standard filler neck. When the use of a standard filler neck is specified during testing, a filler neck conforming to MS35841 shall be used with size A caps and a filler neck conforming to MS51063 shall be used with size B caps.

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select three caps produced under the production contract for first article inspection. First article samples shall be inspected as specified in table I. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply caps that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

TABLE I. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance		
				Examination	Tests	Control
Materials and construction	3.2 and 3.2.1	4.8.1	X			
Defects (see 4.5.2 and table II)	3.2.1, 3.3 thru 3.3.1.2, 3.5 and 3.6	4.8.2	X	X		
Fluid resistance	3.4.1	4.8.3	X			X
Corrosion resistance	3.4.2	4.8.4	X			X
Extreme low temperature	3.4.3	4.8.5	X			X
Pressure valve	3.4.4	4.8.6	X		X	X
Leakage	3.4.4.1	4.8.6.1	X		X	X
Vacuum valve	3.4.5	4.8.7	X		X	X

4.4.1 First article inspection failure. Deficiencies found during, or as a result of, first article inspection shall be cause for rejection of the first article sample until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, first article inspection shall be evidence that all items already produced prior to completion of first article inspection are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government will not accept products until first article inspection is completed to the satisfaction of the Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:29Z
Check the source to verify that this is the current version before use.

4.5 Quality conformance inspection (QCI).

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the caps of one size from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.1.3 Sampling for tests. Samples for tests shall be selected in accordance with level L-6 of MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL's).

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table II.

TABLE II. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Nonconformance in design and construction (see 3.3).	Visual
102	Functionability affecting cap connecting to applicable filler neck (see 3.3.1.1 and 3.3.1.2).	Visual
103	Incorrect dimensions affecting interchangeability (see 3.3).	SIE 1/
104	Damaged (cracked, flaking or split) gaskets (see 3.6).	Visual
105	Faulty workmanship affecting performance (see 3.6).	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:29Z
Check the source to verify that this is the current version before use.

TABLE II. Classification of defects - Continued.

Category	Defect	Method of examination
Minor	AQL 2.5% Defective	
201	Incorrect dimensions not affecting interchangeability (see 3.3).	SIE
202	Improper finish (see 3.2.1).	Visual
203	Improper marking (see 3.5).	Visual
204	Dirty internal parts (see 3.6).	Visual
205	Faulty workmanship affecting appearance (see 3.6).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Tests. Samples selected in accordance with 4.5.1.3 shall be subjected to the quality conformance tests specified in table I, in the order specified in table III, and shall conform to an AQL of 1.0.

4.6 Control tests. Control tests shall be conducted on 3 caps from each lot of 500 units consecutively produced, except that not more than 3 tests shall be performed in a 1-month period, nor less than 1 test in a 1-month period. The caps shall be be identified as to production period and from a production lot which has passed the examination specified in 4.5.2.2 and quality conformance tests specified in 4.5.3. Control tests shall be as specified in table I and conducted in the order specified in table III.

TABLE III. Test schedule.

Sample	Inspection		QCI	Control
1	Low temperature	4.8.5		X
	Pressure valve	4.8.6	X	X
	Leakage	4.8.6.1	X	X
	Vacuum valve	4.8.7	X	X
2	Fluid resistance	4.8.3		X
	Corrosion resistance	4.8.4		X
3	Pressure valve	4.8.6		X
	Leakage	4.8.6.1		X
	Vacuum valve	4.8.7		X

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:29Z
Check the source to verify that this is the current version before use.

4.7 Failure. Failure of any cap to pass any of the specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2 and 3.2.1 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.2.1, 3.3 through 3.3.1.2, 3.5 and 3.6 shall be determined by examination for the defects listed in table II. Examination shall be visual, or by measurement with standard inspection equipment.

4.8.3 Fluid resistance. To determine conformance to 3.4.1, gaskets shall be subjected to test method 6211 of FED-STD-601, using coolants in accordance with 3.4.

4.8.4 Corrosion resistance. To determine conformance to 3.4.2, caps shall be subjected to a salt spray test in accordance with ASTM B117. Subsequently, the cap shall pass the tests specified in 4.8.6, 4.8.6.1 and 4.8.7.

4.8.5 Extreme low temperature. To determine conformance to 3.4.3, the torque, as specified, shall be measured with a torque wrench. Visual and tactile verification using arctic mittens at -65°F shall confirm the removal check. This procedure shall be repeated five times.

4.8.6 Pressure valve. To determine conformance to 3.4.4, the cap shall be connected to the standard filler neck of proper size (see figure 1). The filler neck shall be equipped with an air pump. The complete assembly shall be submerged in water held at 180° $\pm$ 5°F. After a 30-minute stabilization period at 180 $\pm$ 5°F, air shall be applied at a rate of increase not to exceed 5 psi per minute until streaming air bubbles are emerging from the cap assembly. The application of air shall then be stopped and pressure noted when bubbles stop. The pressure shall be recorded.

4.8.6.1 Leakage. To determine conformance to 3.4.4.1, cap shall be connected to the standard filler neck, and air applied in the same manner specified in 4.8.6 until a pressure of 6.25 psi has been reached. The pressure around the valve ports shall be maintained for 1 minute. Continual observation of leakage around the valve ports shall be maintained during this period.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:29Z
Check the source to verify that this is the current version before use.

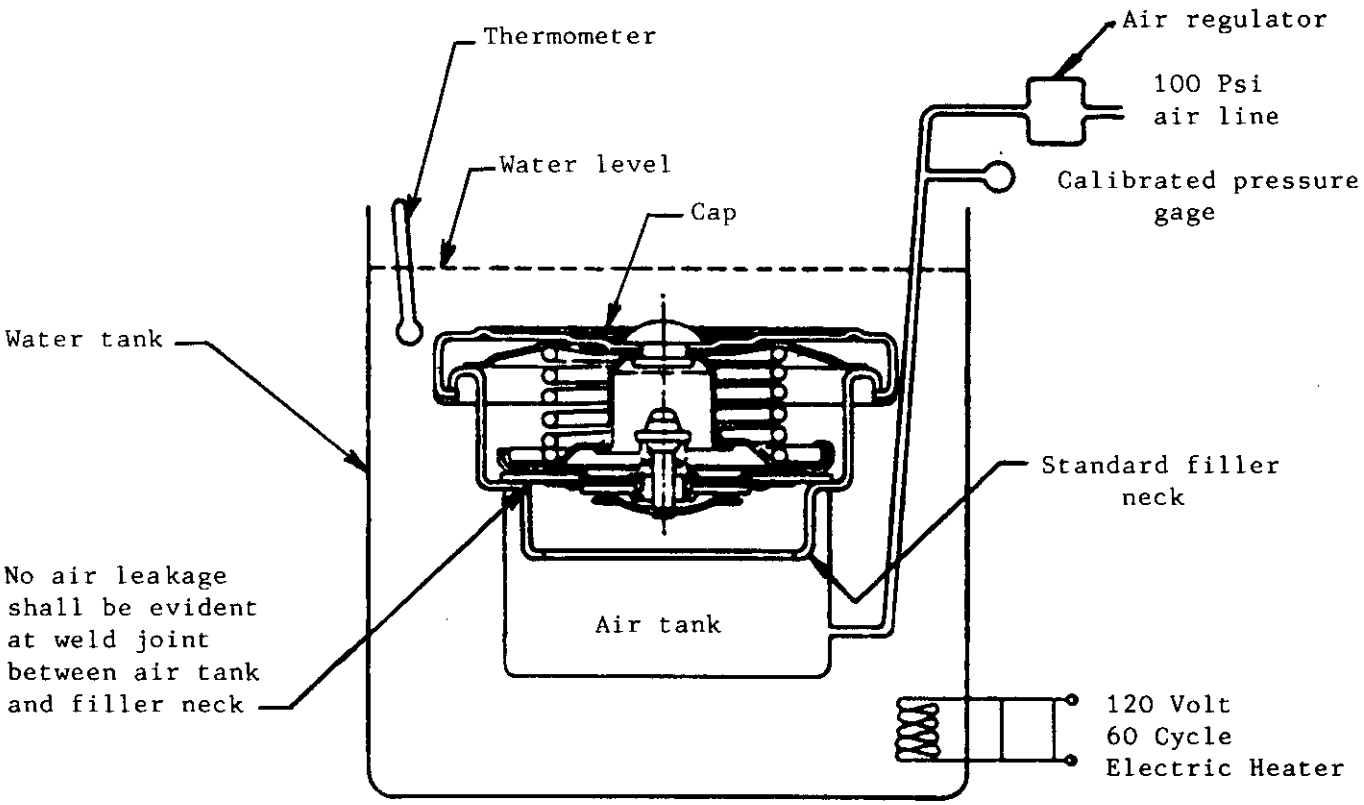


FIGURE 1. Set-up for determining air leakage and opening pressures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:29Z
Check the source to verify that this is the current version before use.

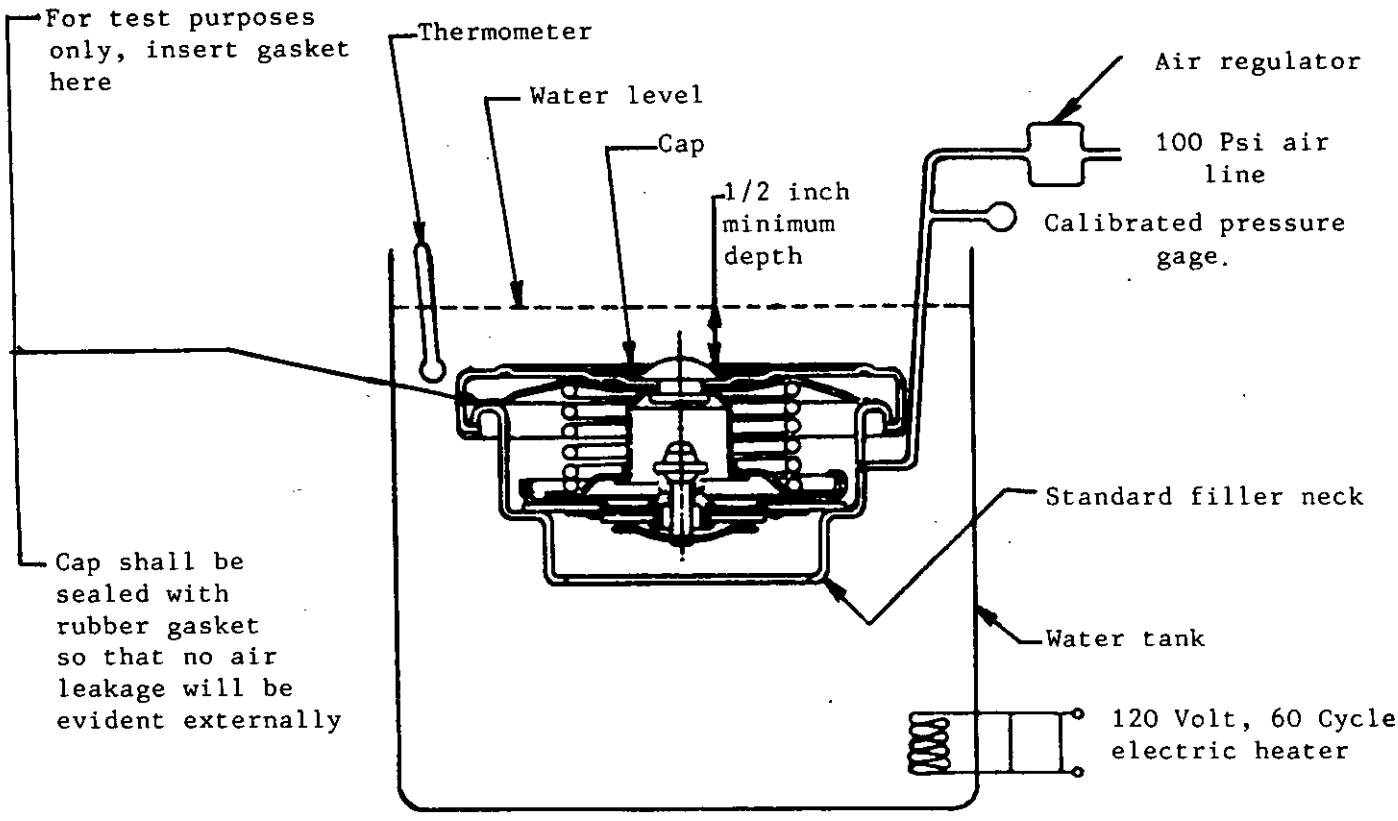


FIGURE 2. Set-up for determining vacuum and opening vacuum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:29Z
Check the source to verify that this is the current version before use.

4.8.7 Vacuum valve. To determine conformance to 3.4.5, the cap shall be connected to a test setup (see figure 2) using a standard filler neck equipped with an air pump. The complete assembly shall be submerged in water maintained at 180 ± 5°F. After a 30-minute stabilization period at 180 ± 5°F, air shall be applied at a rate of increase not to exceed 1 psi per minute until the vacuum valve opens. The opening pressure shall be recorded. As an optional test method, a test setup may be used in which a vacuum is applied to the botton of the cap.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The caps covered by this specification are intended for use on radiators used in cooling systems of industrial and automotive internal-combustion engines.

6.1.1 Size A. This radiator cap is intended for use with a radiator filler neck conforming to MS35841.

6.1.2 Size B. This radiator cap is intended for use with a radiator filler neck conforming to MS51063.

6.2 Ordering data. Acquisition documents should specify the following information:

a. Title, number, and date of this specification.
b. Size of cap required (see 1.2).
c. If first article inspection is not required (see 3.1).
d. If responsibility for inspection shall be other than as specified (see 4.1).
e. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
f. If inspection conditions shall be other than as specified (see 4.3).
g. If first article sample size shall be other than specified (see 4.4).
h. Selection of applicable level and packaging requirements (see 5.1).

6.3 Definitions.

6.3.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.3.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:29Z
Check the source to verify that this is the current version before use.

6.3.2 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.4 Subject term (key word) listing.

Radiator caps, pressure relieving
Pressure relieving radiator caps

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AT
Air Force - 99

Preparing activity:
Army - AT

(Project 2930-0121)

☆U.S. GOVERNMENT PRINTING OFFICE: 1988 - 505-033/92414

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:29Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Tank-Automotive Command
ATTN: AMSTA-GDS
Warren, MI 48397-5000

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:29Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:29Z
Check the source to verify that this is the current version before use.